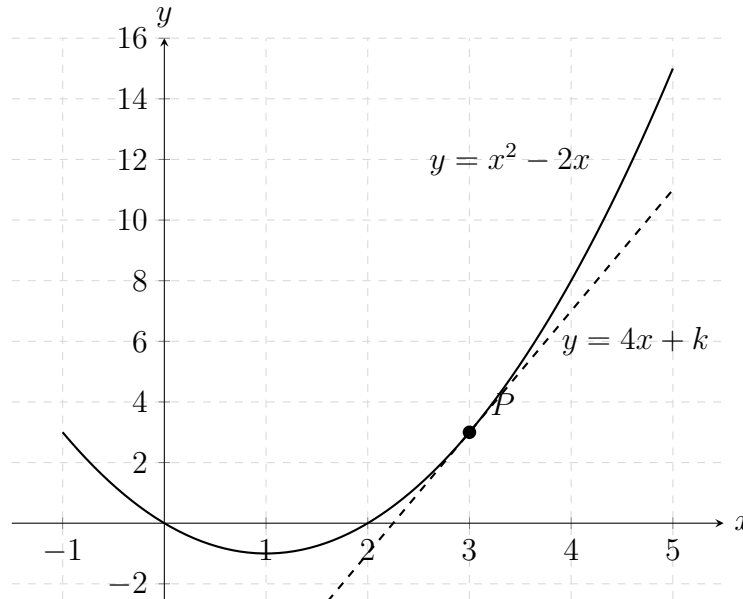


6.2 Classwork: Quadratics

1. The diagram below shows the parabola $y = x^2 - 2x$ and the line $y = 4x + k$, where k is a constant. The line is tangent to the parabola at the point P .



- (a) Write down an equation whose solutions give the x -coordinates of the points where the line meets the parabola. [2]
- (b) Simplify your equation to the form $x^2 + bx + c = 0$, stating b and c in terms of k . [2]
- (c) For the line to be tangent, state the condition on the discriminant. [1]
- (d) Find the value of k . [3]
- (e) Find the coordinates of the point P . [2]

2. Let $f(x) = -x^2 + 4x + 5$.

- Write $f(x)$ in the form $-(x - h)^2 + k$. [3]
- Write down the coordinates of the vertex of the parabola. [1]
- Write down the equation of the axis of symmetry. [1]
- Find the x -intercepts of the graph of f . [2]
- Write down the y -intercept. [1]

3. The equation $2x^2 - 4x + p = 0$ has two distinct real roots, where $p \in \mathbb{R}$.
 - (a) Find the discriminant of the equation in terms of p . [2]
 - (b) Find the values of p for which the equation has two distinct real roots. [2]
 - (c) Find the values of p for which the equation has no real roots. [1]
 - (d) Find the value of p for which there is exactly one real root, and find that root. [3]

